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## Avoiding CAN Remote Frames in CANopen networks

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May 10, 2021



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This version replaces Version 1.1.0, which has been revised editorially. The main changes are as follows:

- Adding of Foreword and Introduction sections;
- Improving wording including using inclusive language; and
- Providing other minor editorial improvements.

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## Introduction

This document recommends not using CAN Remote Frames, because the re-action on received CAN Remote Frames is implementation-specific. Additionally, the user needs to provide the data length code (DLC) of the requested CAN Data Frame in the transmitted CAN Remote Frame. In order to reduce busload and to avoid unexpected behavior, it is recommended to map CANopen communication services only to CAN Data Frames.

## CONTENTS

|       |                                                                  |    |
|-------|------------------------------------------------------------------|----|
| 1     | Scope .....                                                      | 5  |
| 2     | Normative references .....                                       | 5  |
| 3     | Terms and definitions .....                                      | 5  |
| 4     | Symbols and abbreviated terms .....                              | 5  |
| 5     | CAN Remote Frame .....                                           | 5  |
| 5.1   | General .....                                                    | 5  |
| 5.2   | Usage problems .....                                             | 6  |
| 5.3   | Usage avoidance and substitution by other CANopen services ..... | 7  |
| 5.3.1 | General .....                                                    | 7  |
| 5.3.2 | Network management .....                                         | 7  |
| 5.3.3 | Handling of the PDOs .....                                       | 9  |
| 5.3.4 | Handling of MPDOs .....                                          | 10 |

## 1 Scope

This document provides recommendations for avoiding CAN Remote Frames when using CANopen communication services.

## 2 Normative references

The following referenced documents are indispensable for the application of this document. For undated references, the latest edition of the referenced document (including any amendments) applies.

CiA 301, CANopen Application layer and communication profile

CiA 302-2, CANopen Additional application layer functions — Part 2: Network management

ISO 11898-1, Road vehicles — Controller area network (CAN) — Part 1: Data link layer and physical signalling

## 3 Terms and definitions

CiA, IEC, and ISO maintain terminological databases for use in specifications respectively standards at the following addresses:

- CiA glossary of terms: <https://www.can-cia.org/groups/cia-glossary-of-terms/>
- IEC Electropedia: <http://www.electropedia.org/>
- ISO Online browsing platform: <https://www.iso.org/obp/>

For the purpose of this document, the terms and definitions given in CiA 301, CiA 302-2, and ISO 11898-1 apply.

## 4 Symbols and abbreviated terms

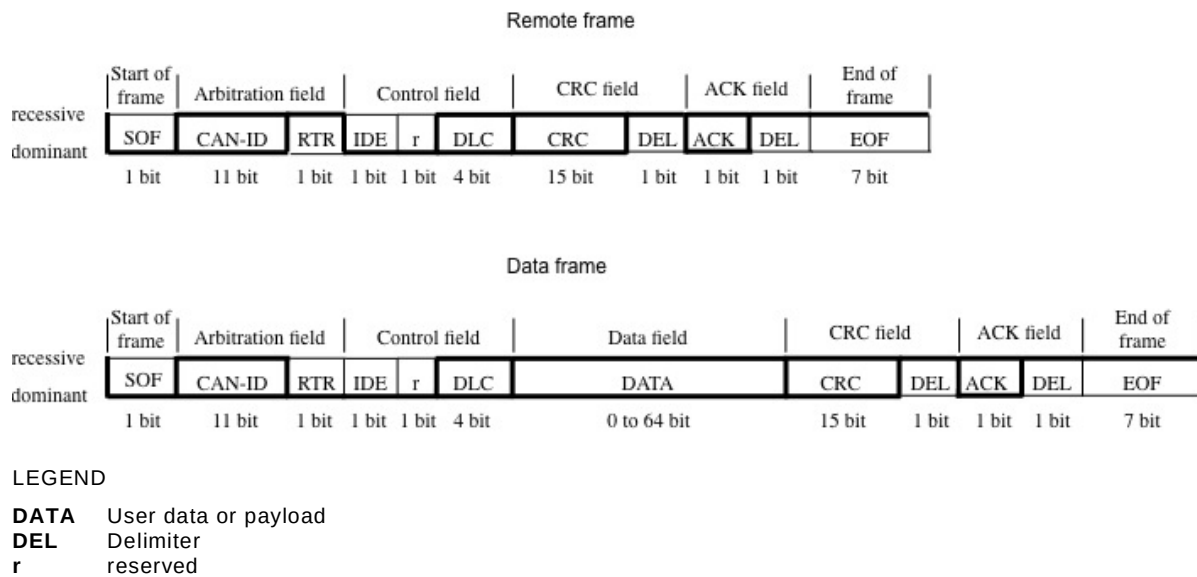
For the purpose of this document, the symbols and abbreviated terms given in CiA 301, CiA 302-2, and ISO 11898-1 apply.

## 5 CAN Remote Frame

### 5.1 General

The CAN data link layer services as specified in ISO 11898-1 include the CAN Data Frame remote request by means of the corresponding CAN Remote Frame. Figure 1 shows the CAN Remote Frame protocol. The CAN Remote Frame is structured in the same way as the CAN Data Frame. The CAN Remote Frame differs from the CAN Data Frame as follows:

- The RTR bit is of recessive value;
- The data field does not exist at all.



**Figure 1 – CAN Data Frame and the CAN Remote Frame formats with an 11-bit CAN-ID**

The CAN Remote Frame requests the transmission of the corresponding CAN Data Frame with the very same CAN identifier (CAN-ID). Due to the fact that the RTR bit is always dominant in CAN Data Frames, the CAN Data Frame wins bus arbitration in the case that one or more CAN Remote Frame with the very same CAN-ID are transmitted at the very same moment.

The CAN data link layer as specified in ISO 11898-1 does not specify the implementation of CAN Remote Frames. There are two main implementations on the market:

- The CAN controller responds to CAN Remote Frames immediately and independent of the micro-controller;
- The CAN controller passes the CAN Remote Frame by means of an interrupt to the host-controller, which sends the corresponding CAN Data Frame.

Some CAN controller implements both behaviors, which are selectable by the node or device designer.

## 5.2 Usage problems

Several general usage problems occur in some CAN controllers during processing of the CAN Remote Frames. These problems are listed as follows:

- The number of transmit messages that support CAN Remote Frames is limited due to the fact that the global receive buffer in some CAN controllers are not able to handle all RTR-requests, therefore only a few RTR-messages are handled. (e.g. up to 14 or sometimes even less RTR-messages are processed);
- The implementation effort of the CAN software driver is more complex because the CAN Remote Frame is relatively complex to handle in some CAN controllers. In some CAN controllers with single message buffering capability, CAN Remote Frames are transmitted via a receive buffer and not via a transmit buffer. If there are more CAN Remote Frames to be handled than receive buffers available, the CAN Remote Frame is either received in the buffer, which transmitted the CAN Remote Frame or received in the global receive buffer;
- When two or more nodes request the CAN Data Frame at the very same moment, a bus collision occurs when the data length codes are different as for arbitration only the CAN identifier bits and the RTR bit are used. From this it follows, e.g. when two or more nodes request the same CAN Data Frame, all receive message buffers linked to this CAN Data Frame needs to have the same data length,

- Furthermore, some CAN controllers with single frame buffer capability respond to a CAN Remote Frame with the corresponding CAN Data Frame with the same data length code received by the CAN Remote Frame and not with originally configured data length;
- In case the application crashes, some CAN controller still answers CAN Remote Frames with old data.

### 5.3 Usage avoidance and substitution by other CANopen services

#### 5.3.1 General

From the beginning, the usage of CAN Remote Frame was restricted in CANopen. CAN Remote Frames are only allowed in the error control service, the node-guarding function, and for remotely requested PDOs.

#### 5.3.2 Network management

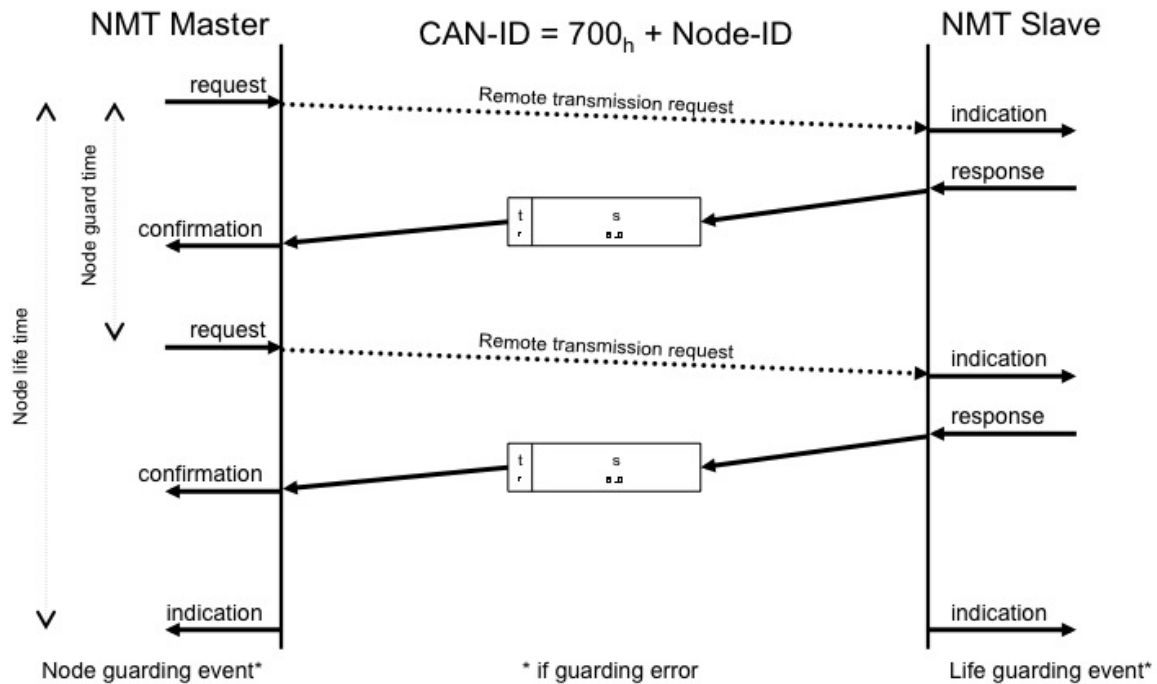
It is recommended to use the Heartbeat mechanism to supervise CANopen nodes with NMT consumer functionality.

With the node-guarding mechanism the CANopen node with NMT producer functionality supervises all NMT consumer nodes in the network. Therefore, the NMT producing node transmitting periodically the node-guarding request by means of a CAN Remote Frame to each NMT consuming node, which responds with its corresponding node-guarding response mapped to a CAN Data Frame (see Figure 2).

The node-guarding response message contains the NMT FSA (finite state automaton) status information plus an additional toggle-bit. The toggle-bit is necessary to detect that the micro-controller is not more updating the status information, when the CAN controller responds automatically without interference of the micro-controller. The node-guarding mechanism has the disadvantage that only the node with NMT producer functionality receives the node-guarding messages. The further disadvantage is that after sending the CANopen boot-up message, the transmit buffer on CAN controller chips with single frame buffers needs to be reloaded with the new data for node-guarding. When however the node-guarding message is already requested by the NMT producing node by means of a CAN Remote Frame before the transmit buffer is updated, the previous buffer transmits the CANopen boot-up message again.

Due to these disadvantages of the node-guarding mechanism, the CiA 301 version 4.0 introduced the heartbeat mechanism. With the heartbeat mechanism any CANopen node is able to supervise other CANopen nodes in the network (see Figure 3). The heartbeat message contains the current NMT FSA state but without the toggle bit. The CANopen node with NMT producer functionality supervises all CANopen NMT consumer nodes in the network using the heartbeat. In this case, the NMT consumer nodes (act as heartbeat producer) send periodically their heartbeat message as CAN Data Frame and the NMT producer node (acts as heartbeat consumer) receives these heartbeat messages (see CiA 301). The NMT producer node checks whether the heartbeat from each node is received within the time interval, configured for this CANopen node in the heartbeat consumer object (see CiA 301). If it is not the case, the heartbeat event occurs. This event indicates that the node is not capable for communication in the network. In the same way one or more CANopen NMT consumer nodes can monitor the condition of CANopen node with NMT producer capability or other CANopen NMT consumer nodes. CANopen consumer nodes with NMT producer functionality also monitor the heartbeat of the active NMT producer node in order to detect the NMT producer failure (see flying NMT producer procedure in CiA 302-2).

The configured heartbeat consumer time should be higher than the heartbeat producer time, in order to ensure the reception of the heartbeat message in time, even in case the time jitters occur.



#### LEGEND

**s:** State of the NMT consumer node

4: Stopped

5: Operational

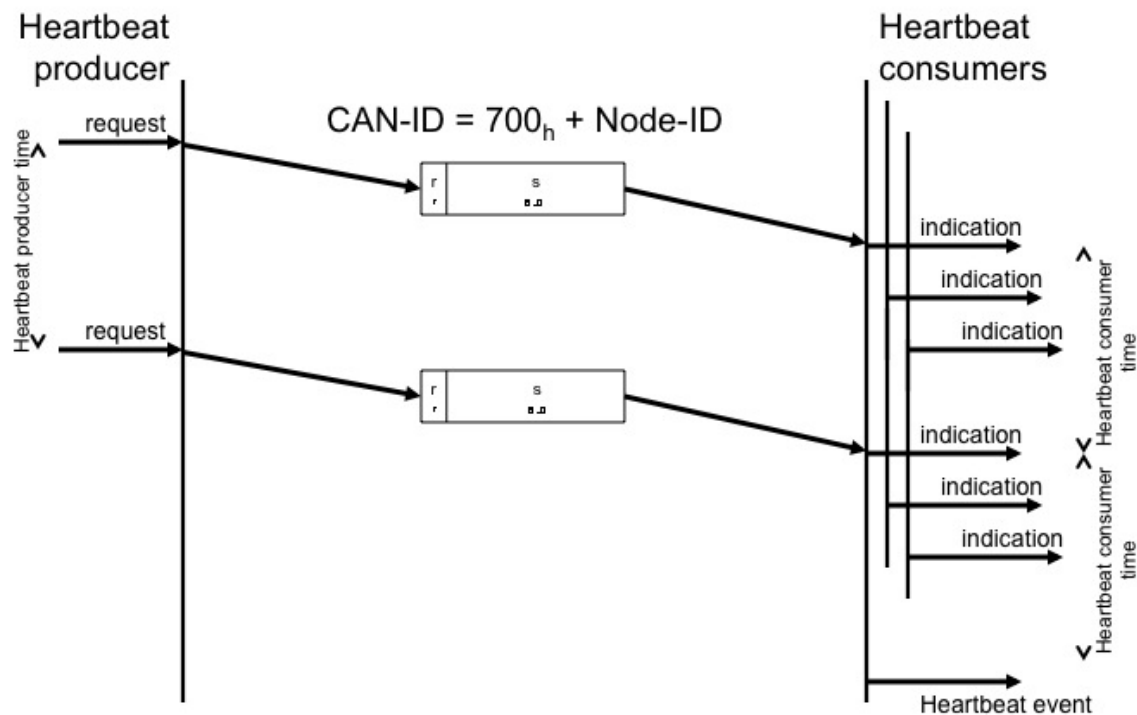
127: Pre-operational

**t:** Toggle bit

**NOTE** The value of the toggle bit alternates between two consecutive responses from the NMT consumer node. The value of the toggle-bit of the first response after the guarding protocol becomes active is 0. The toggle bit in the guarding protocol is reset to 0 when the NMT sub-state reset communication is passed (no other change of NMT state resets the toggle bit). If a response is received with the same value of the toggle-bit as in the preceding response then the new response is handled as if it was not received.

**Figure 2 – Node guarding protocol**





#### LEGEND

- r: reserved (always 0)
- s: State of the heartbeat producer
  - 0: Boot-up
  - 4: Stopped
  - 5: Operational
  - 127: Pre-operational

**Figure 3 – Heartbeat protocol**

The advantages of the heartbeat in comparison to the node-guarding protocol are listed as follows:

- Reduced busload as no CAN Remote Frame is necessary;
- CANopen NMT consumer nodes monitor directly other CANopen nodes (faster reaction time compared to the NMT producer approach, reporting node failures to the remaining NMT consumer nodes);
- Decentralized monitoring is better than centralized monitoring in case the node with NMT producer functionality fails.

### 5.3.3 Handling of the PDOs

#### 5.3.3.1 General

In particular in PLC-based control systems, the transmission of PDOs should be scheduled by the progression of time in defined frequency. CANopen provides several scheduling mechanism to achieve such behavior. It is not recommended to use CAN Remote Frames for this purpose.

#### 5.3.3.2 Centralized scheduling of PDOs

If a centralized scheduling of PDOs is required, it is recommended to use synchronous PDOs. The Sync message is than the trigger for updating the mapped objects as well as for the PDO transmission. The system designer has the possibility to define that the synchronous PDO serves all received Sync messages or only every second, every third etc. The lowest frequency is to serve every 240<sup>th</sup> Sync message.

In order to reduce busloads, the system designer can use the acyclic transmission of PDOs. Acyclic transmission means that in addition to the reception of the Sync message an internal event (e.g. change of state) has to be occurred in order to cause a PDO transmission.

### **5.3.3.3 Decentralized scheduling of PDO**

If a decentralized scheduling of PDOs is required, the event-timer should be used to trigger PDOs. A typical application for requesting PDOs by CAN Remote Frames would be when a newly connected HMI (human machine interface) device immediately retrieves all necessary process data from the other CANopen NMT consumer nodes. This is especially important if the PDOs are only transmitted “on change” (event-triggered). This can take some time.

In order to retrieve the process data on one side but to avoid the usage of CAN Remote Frames on the other side, the PDO event timer should be used. With this event timer individual time is configurable for each TPDO. The PDO is re-transmitted, even if the corresponding process data has not changed.

### **5.3.4 Handling of MPDOs**

According to CiA 301, MPDOs are not allowed to be requested remotely by means of CAN Remote Frames.